

Recall: Invariant Subspace of $T \in \mathcal{L}(V)$.

$$U \leq V \quad \text{and} \quad T(U) \subseteq U.$$

Then we say U is invariant under T
(T -invariant).

?: What is the matrix of T under a basis of V when $U \leq V$ is T -invariant?

ANS: Take $u_1, \dots, u_k$ a basis for U .

By Basis Extension, we can find $u_{k+1}, \dots, u_n$

such that $(\underbrace{u_1, \dots, u_k}_{\substack{\text{basis of } U \\ T\text{-invariant}}}, u_{k+1}, \dots, u_n)$ is a basis of V .

$$T(u_1, \dots, u_k, u_{k+1}, \dots, u_n) = (\underbrace{u_1, \dots, u_k}_{\substack{\text{basis of } U \\ T\text{-invariant}}}, u_{k+1}, \dots, u_n) \begin{bmatrix} A_{k \times k} & C \\ 0 & B \end{bmatrix}$$

As a corollary:

If $V = U \oplus W$ and both U and W are T -invariant. Then the matrix of T under basis

$$(\underbrace{u_1, \dots, u_k}_U, \underbrace{w_1, \dots, w_j}_W) \text{ is } \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}$$

If this is unclear, try writing out

$$T(u_1) = a_{11}u_1 + a_{21}u_2 + \dots + a_{k1}u_k + 0w_1 + \dots + 0w_j$$

$$T(u_2) = a_{12}u_1 + a_{22}u_2 + \dots + a_{k2}u_k + 0w_1 + \dots + 0w_j$$

$\vdots$

$$T(u_k) = 0u_1 + 0u_2 + \dots + 0u_k + b_{1j}w_1 + b_{2j}w_2 + \dots + b_{jj}w_j$$

$\vdots$

$$T(w_j) = 0u_1 + 0u_2 + \dots + 0u_k + b_{1j}w_1 + b_{2j}w_2 + \dots + b_{jj}w_j$$

Collect this in matrix form we have

$$T(u_1, \dots, u_k, w_1, \dots, w_j) = (u_1, \dots, u_k, w_1, \dots, w_j) \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}$$

Q: What is the simplest invariant subspace for a linear transformation? i.e. one-dimensional T -invariant subspace.

Note: A one dimensional space is $\text{Span}(v)$.



Ans: For $\text{Span}(v) \leq V$, to be T -invariant.

It has to be that $T(v) = \lambda v$ for some $\lambda \in K$.
This leads to the concept of eigenvalues and eigenvectors.

- Eigenvalues and Eigenvectors.

Note: For this section, we require that the field K is algebraically closed, meaning that any polynomial with coefficients in K has a root in K .

- $\mathbb{Q}$ is a field, NOT algebraically closed.

e.g. $x^2 - 2 = 0 \Rightarrow x = \pm\sqrt{2} \notin \mathbb{Q}$

- $\mathbb{R}$ is a field, NOT algebraically closed.

e.g. $x^2 + 1 = 0 \Rightarrow x = \pm i \notin \mathbb{R}$

- $\mathbb{C}$ is a field, IS algebraically closed.

If you want, take $K = \mathbb{C}$.

- Definition: $(V, +, \cdot)$, $T \in \mathcal{L}(V)$.

If $v \neq 0_V$: $T(v) = \lambda v$ for some $\lambda \in K$.

We call λ an eigenvalue of T and v its associated eigenvector.

! : Eigenvectors has to be non-zero.

Definition: $(K^n, +, \cdot)$, $A \in K^{n \times n}$

$x \neq 0_{K^n}$: $\underline{Ax = \lambda x}$ for some $\lambda \in K$
 λ an eigenvalue of A
 x its eigenvector.

- Key condition for eigenvalue/vectors:

$$T \in \mathcal{L}(V)$$

$$v \neq 0_v : T(v) = \lambda v (= \lambda \text{id}_v(v))$$

$$\Leftrightarrow (T - \lambda \text{id}_v)(v) = 0_v$$

$$\Leftrightarrow E_\lambda(T) := \ker(T - \lambda \text{id}_v) \neq \{0_v\}$$

$\Leftrightarrow T - \lambda \text{id}_v$ is NOT injective

$\Leftrightarrow T - \lambda \text{id}_v$ is NOT surjective
 ----- " NOT bijection

$\Leftrightarrow T - \lambda \text{id}_v$ is NOT invertible



$$\det(T - \lambda \text{id}_v) := \det(A - \lambda I_n) = 0$$

$$A \in K^{n \times n}$$

$$x \neq 0_{K^n} : Ax = \lambda x (= \lambda I_n x)$$

$$\Leftrightarrow (A - \lambda I_n)x = 0_{K^n}$$

$$\Leftrightarrow E_\lambda(A) := \ker(A - \lambda I_n) \neq \{0_{K^n}\}$$

$\Leftrightarrow A - \lambda I_n$ is NOT injective
 (viewed as $\mathcal{L}(K^n)$)

$\Leftrightarrow A - \lambda I_n$ is NOT surjective
 ----- " NOT bijection

$\Leftrightarrow A - \lambda I_n$ is NOT invertible.



- Procedure to find eigenvalue/vectors for $A \in K^n$



for $T \in \mathcal{L}(V)$ whose matrix
is $A \in K^n$ under a basis.

1. Calculate $\det(A - \lambda I_n)$.

2. Set $p_A(\lambda) := \det(A - \lambda I_n) = 0$ solve for its roots.

$\lambda_1, \dots, \lambda_n$ (some λ_i s might be the same)

3. For every distinct λ_i , solve linear system

$$(A - \lambda_i I_n) x = 0_{K^n}$$

$\swarrow$ eigenspace

to get eigenvectors (in $\ker(A - \lambda_i I_n) =: E_{\lambda_i}(A)$).

- Basic properties of Eigenvectors and Eigenspaces.

- Claim: Eigenvectors associated with distinct eigenvalues are linearly independent.

Proof: Take $v_1, \dots, v_s$ to be the eigenvectors for $\lambda_1, \dots, \lambda_s$ and $\lambda_i \neq \lambda_j$ for all $i \neq j$.

Suppose $v_1, \dots, v_s$ is lin. dep.

△ Define $1 \leq m \leq s$ to be the minimal integer such that

→ $C_1 v_1 + C_2 v_2 + \dots + C_m v_m = 0_v$ has nontrivial C_i s.

$$\begin{aligned} T(0_v) &= T(C_1 v_1 + C_2 v_2 + \dots + C_m v_m) = C_1 T(v_1) + C_2 T(v_2) + \dots + C_m T(v_m) \\ &= C_1 \lambda_1 v_1 + C_2 \lambda_2 v_2 + \dots + C_m \lambda_m v_m \\ &= 0_v \end{aligned}$$

$$C_1 \lambda_m v_1 + C_2 \lambda_m v_2 + \dots + C_m \lambda_m v_m = 0_v \quad (\times \lambda_m)$$

$$\text{⊖: } C_1 \lambda_1 v_1 + C_2 \lambda_2 v_2 + \dots + C_m \lambda_m v_m = 0_v$$

$$C_1 (\lambda_m - \lambda_1) v_1 + C_2 (\lambda_m - \lambda_2) v_2 + \dots + C_{m-1} (\lambda_m - \lambda_{m-1}) v_{m-1} = 0_v \quad (*)$$

In $C_1, \dots, C_m$, there must be at least 2 nonzero numbers because v_i s are eigenvectors

Then (*) finds $m-1$ coefficients, nontrivial such that the linear combination of $v_1, \dots, v_{m-1} = 0_v$.

This contradicts the minimality of m .

□

- Claim: If λ, μ eigenvalues and $\lambda \neq \mu$.

$$E_\lambda(T) \cap E_\mu(T) = \{0_V\}$$

Proof: Take $v \in E_\lambda(T) \cap E_\mu(T)$

$$\text{then } T(v) = \lambda v = T(v) = \mu v$$

$$\text{thus } (\lambda - \mu)v = 0_V \Rightarrow v = 0_V \text{ since } \lambda \neq \mu.$$

□

- Note: Eigenspace is all eigenvectors $\cup$ the zero vector.

- Claim: Eigenspace is a subspace. ($\ker(T - \lambda \text{id}_V)$.)

Eigenspace is T -invariant. (Easy check).

- Diagonalization:

Suppose we have n eigenvectors $v_1, \dots, v_n$ for eigenvalues $\lambda_1, \dots, \lambda_n$ (some might be the same) for $T \in \mathcal{L}(V)$

$v_1, \dots, v_n$ lin. indep. and since $\dim V = n$.

$v_1, \dots, v_n$ form a basis for V .

Then

$$T(v_1, \dots, v_n) = (v_1, \dots, v_n) \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

$$= (v_1, \dots, v_n) \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}.$$

We call that T is diagonalizable.

In the matrix world: $A \in K^{n \times n}$ $p_1, \dots, p_n \in K^n$ eigenvectors
lin. indep.

$$A \begin{pmatrix} | & | & & | \\ p_1 & p_2 & \dots & p_n \\ | & | & & | \end{pmatrix} = \begin{pmatrix} | & | & & | \\ p_1 & p_2 & \dots & p_n \\ | & | & & | \end{pmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \ddots & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}$$

Let $P = \begin{bmatrix} | & | & & | \\ p_1 & p_2 & \dots & p_n \\ | & | & & | \end{bmatrix}$ since $p_1, p_2, \dots, p_n$ are basis

P has full rank $\Leftrightarrow P$ invertible.

$$P^{-1}AP = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}.$$

We say A is diagonalizable.

• The key for $T \in \mathcal{L}(V)$ ($A \in K^{n \times n}$) to be diagonalizable is that we can find a basis for V from its eigenvectors.

• Remark: Not all matrices are diagonalizable!

(See assignment 1).

- Why is diagonalization useful?

One reason: Calculating A^k direct computation is lengthy.

If A is diagonalizable. Then $\exists P$ invertible.

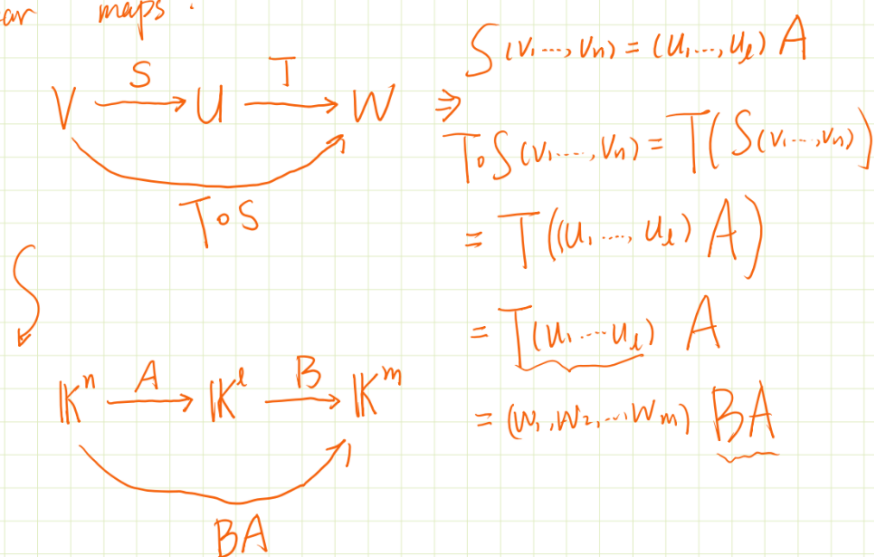
$$D = P^{-1}AP = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \Leftrightarrow A = PDP^{-1} \text{ where } D = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$$

$$\text{Then } A^k = (PDP^{-1})^k = \underbrace{(PDP^{-1})}_{I_n} \underbrace{(PDP^{-1})}_{I_n} \dots \underbrace{(PDP^{-1})}_{I_n}$$

$$= PD^kP^{-1} = P \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}^k P^{-1}$$
$$= P \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix} P^{-1}.$$

- A Few Notes on Matrix Multiplication.

- Matrix Multiplication represents composition of linear maps:



- Four ways of looking at $A_{m \times l} B_{l \times n}$.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1l} \\ a_{21} & a_{22} & \dots & a_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{ml} \end{pmatrix} = \begin{pmatrix} \text{--- } a_1^T \text{ ---} \\ \text{--- } a_2^T \text{ ---} \\ \vdots \\ \text{--- } a_m^T \text{ ---} \end{pmatrix} \quad a_i \in K^l$$

$$B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{l1} & b_{l2} & \dots & b_{ln} \end{pmatrix} = \begin{pmatrix} | & | & | & \dots & | \\ b_1 & b_2 & b_3 & \dots & b_n \\ | & | & | & \dots & | \end{pmatrix} \quad b_i \in K^l$$

- Element-wise: Let $C = AB$

$$c_{ij} = \sum_{k=1}^l a_{ik} b_{kj} \quad i \left(\text{---} \right) \left(\begin{array}{c} i \\ | \end{array} \right)$$

- Matrix by columns: $A_{m \times l} b_{l \times 1} = m \times 1$

$$AB = A(b_1, \dots, b_n) = (Ab_1, Ab_2, \dots, Ab_n)_{m \times n}$$

- Rows by Matrix:

$$AB = \begin{pmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_m^T \end{pmatrix} B = \begin{pmatrix} a_1^T B \\ a_2^T B \\ \vdots \\ a_m^T B \end{pmatrix}_{m \times n} \quad \begin{matrix} 1 \times l & l \times n \end{matrix} = 1 \times n$$

- Inner product view:

$$AB = \begin{pmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_m^T \end{pmatrix} (b_1, \dots, b_n) = \begin{pmatrix} \overbrace{a_1^T b_1, a_1^T b_2, \dots, a_1^T b_n}^{e \in K} \\ \vdots \\ a_m^T b_1, a_m^T b_2, \dots, a_m^T b_n \end{pmatrix}$$

- Four ways of looking at $A_{m \times l} B_{l \times n}$.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1l} \\ a_{21} & a_{22} & \dots & a_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{ml} \end{pmatrix} = \begin{pmatrix} \text{---} a_1 \text{---} \\ \text{---} a_2 \text{---} \\ \vdots \\ \text{---} a_m \text{---} \end{pmatrix} \quad a_i \in K^l$$

$$B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{l1} & b_{l2} & \dots & b_{ln} \end{pmatrix} = \begin{pmatrix} | & | & | & \dots & | \\ b_1 & b_2 & b_3 & \dots & b_n \\ | & | & | & \dots & | \end{pmatrix} \quad b_i \in K^l$$

1. Element-wise: let $C = AB$

$$c_{ij} = \sum_{k=1}^l a_{ik} b_{kj} \quad i \left(\text{---} \right) \left(\begin{matrix} j \\ | \end{matrix} \right)$$

2. Matrix by columns: $A_{m \times l} b_{l \times 1} = m \times 1$

$$AB = \underset{m \times l \quad l \times n}{A} (b_1, \dots, b_n) = (Ab_1, Ab_2, \dots, Ab_n)_{m \times n}$$

3. Rows by Matrix:

$$AB = \begin{pmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_m^T \end{pmatrix} B = \begin{pmatrix} a_1^T B \\ a_2^T B \\ \vdots \\ a_m^T B \end{pmatrix}_{m \times n} \quad \underset{1 \times l \quad l \times n}{a_i^T B} = 1 \times n$$

4. Inner product view:

$$AB = \begin{pmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_m^T \end{pmatrix} (b_1, \dots, b_n) = \begin{pmatrix} \overset{\in K}{a_1^T b_1}, a_1^T b_2, \dots, a_1^T b_n \\ \vdots \\ a_m^T b_1, a_m^T b_2, \dots, a_m^T b_n \end{pmatrix}$$

- Inner Product Spaces

• Motivation:

So far for $(V, +, \cdot)$ we can $+$, $\cdot$, choose basis and study linear maps (transformations).

Missing: Length of a vector.
 Angle between two vectors. } These are available in $\mathbb{R}^2$ and $\mathbb{R}^3$.

• Review: Dot product in $\mathbb{R}^n$.

$$x = (x_1, \dots, x_n)^T, \quad y = (y_1, \dots, y_n)^T$$

$$x \cdot y = x_1 y_1 + x_2 y_2 + \dots + x_n y_n = x^T y = (x_1, \dots, x_n) \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$x \cdot x = x_1^2 + x_2^2 + \dots + x_n^2 = \|x\|^2 \text{ (square of length)}$$

If $x \cdot y = 0$, we know that $x \perp y$.

The concept of Inner Product Spaces generalizes this.

- Definition:

Let V be a vec. space over $\mathbb{R}$ or $\mathbb{C}$.

An inner product on V is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow K (= \mathbb{R} \text{ or } \mathbb{C})$$

Such that for all $u, v, w \in V$ and $\alpha, \beta \in K$

(1). Linearity in the first argument:

$$\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$$

(2). Conjugate symmetry:

$$\langle u, v \rangle = \overline{\langle v, u \rangle} \quad \left(\begin{array}{l} \text{Recall if } z \in \mathbb{C}, z = a + bi, a, b \in \mathbb{R} \\ \text{Conjugate of } z : \bar{z} = a - bi, a, b \in \mathbb{R} \end{array} \right)$$

▷ (3). Positive Definiteness:

$$\langle v, v \rangle \geq 0 \text{ with } \langle v, v \rangle = 0 \Leftrightarrow v = 0,$$

A vector space equipped with an inner product is called an inner product space.

$(V, +, \cdot) \xrightarrow{\text{add } \langle \cdot, \cdot \rangle} (V, +, \cdot, \langle \cdot, \cdot \rangle)$ an inner product space.

Note: If $K = \mathbb{R}$. then conjugate symmetry

reduced to symmetry because if $x \in \mathbb{R}$. $\bar{x} = x$.

$\langle u, v \rangle = \langle v, u \rangle$ for real inner product spaces.

Examples of inner product spaces:

1. $\mathbb{R}^n$ with standard dot product:

$$\langle x, y \rangle = x \cdot y = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

2. $\mathbb{R}^n$ with an alternative inner prod.

Take $c_1, c_2, \dots, c_n \in \mathbb{R}$ $c_i > 0, \forall i$

$$\langle x, y \rangle = c_1 x_1 y_1 + c_2 x_2 y_2 + \dots + c_n x_n y_n$$

3. $C[a, b]$ = Space of continuous real-valued functions on $[a, b]$.

$$\langle f, g \rangle = \int_a^b f(t)g(t) dt$$

- Norm (Length) induced by an inner product:

$$\text{For } v \in V : \|v\| = \sqrt{\langle v, v \rangle}$$

Example: for $\mathbb{R}^n$ take $\cdot$:

$$\|x\| = \sqrt{x \cdot x} = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

- Properties of a Norm:

For every $v \in V$, and $\alpha \in K (= \mathbb{R} \text{ or } \mathbb{C})$

We have: $\|v\| \geq 0$ and $\|v\| = 0 \Leftrightarrow v = 0$. (1)

$$\langle v, v \rangle \geq 0 \\ \alpha \cdot v \in \mathbb{R}$$

$$\text{and } \|\alpha v\| = |\alpha| \|v\|. \quad (2)$$

Note: (1). Follows from positive definiteness of $\langle \cdot, \cdot \rangle$

$$\begin{aligned} (2). \|\alpha v\|^2 &= \langle \alpha v, \alpha v \rangle = \alpha \langle v, \alpha v \rangle, \quad \overline{zw} = \overline{z} \overline{w} \quad z, w \in \mathbb{C} \\ &= \alpha \overline{\langle \alpha v, v \rangle} = \alpha \overline{\alpha \langle v, v \rangle} = \alpha \overline{\alpha} \overline{\langle v, v \rangle} \\ &= \alpha \overline{\alpha} \langle v, v \rangle = |\alpha|^2 \|v\|^2 \end{aligned}$$

□

- Perpendicularity: Define $u \perp v$ if $\langle u, v \rangle = 0$

$$\Rightarrow \langle v, u \rangle = \overline{\langle u, v \rangle} = \overline{0} = 0 \Rightarrow v \perp u$$



- Pythagorean Theorem:

If $u \perp v$, then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$.



Proof: Since $u \perp v$: $\langle u, v \rangle = \langle v, u \rangle = 0$. then

$$\|u+v\|^2 = \langle u+v, u+v \rangle = \langle u, u+v \rangle + \langle v, u+v \rangle$$

$$= \overline{\langle u+v, u \rangle} + \overline{\langle u+v, v \rangle}$$

$$= \overline{\langle u, u \rangle + \langle v, u \rangle} + \overline{\langle u, v \rangle + \langle v, v \rangle}$$

$$\begin{matrix} 1 \\ 0 \end{matrix}$$

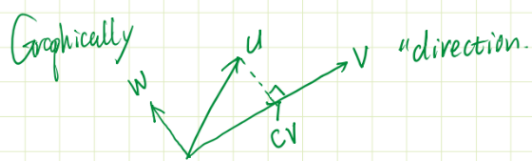
$$\begin{matrix} 1 \\ 0 \end{matrix}$$

$$= \|u\|^2 + \|v\|^2$$

- Decomposing a vector along a direction.

Given $v \neq 0$, the direction.

How can we write $u \in V$: $u = cv + w$, $w \perp v$



Since we want $w = u - cv \perp v$

$$0 = \langle u - cv, v \rangle = \langle u, v \rangle - c \langle v, v \rangle = \langle u, v \rangle - c \|v\|^2$$

Hence $c = \frac{\langle u, v \rangle}{\|v\|^2}$

Now we have Let $u, v \in V$, $v \neq 0$

$$u = \frac{\langle u, v \rangle}{\|v\|^2} v + w \quad \text{where } w = u - \frac{\langle u, v \rangle}{\|v\|^2} v \text{ and } w \perp v$$

- Cauchy-Schwarz Inequality:

For all $u, v \in V$.

$$|\langle u, v \rangle| \leq \|u\| \|v\| \quad "=" \text{ iff } u, v \text{ are lin. dep.}$$

$u = kv \text{ for some } k \in \mathbb{K}$

Proof: 1°. If $v = 0$, Then $\checkmark$

2°. $v \neq 0$. Then $u = \frac{\langle u, v \rangle}{\|v\|^2} v + w$, $w \perp v$

By Pythagorean Thm.



$$\|u\|^2 = \left\| \frac{\langle u, v \rangle}{\|v\|^2} v \right\|^2 + \|w\|^2$$

$$= \frac{|\langle u, v \rangle|^2}{\|v\|^4} \|v\|^2 + \|w\|^2$$

$$= \frac{|\langle u, v \rangle|^2}{\|v\|^2} + \underbrace{\|w\|^2}_{\geq 0} \geq \frac{|\langle u, v \rangle|^2}{\underbrace{\|v\|^2}_{> 0}}$$

$\|v\| \|u\|^2 \geq |\langle u, v \rangle|^2$ then take square root.

Equality holds iff $\|w\| = 0 \Leftrightarrow w = 0$

Then $u = \frac{\langle u, v \rangle}{\|v\|^2} v$

□

For $K = \mathbb{R}$:

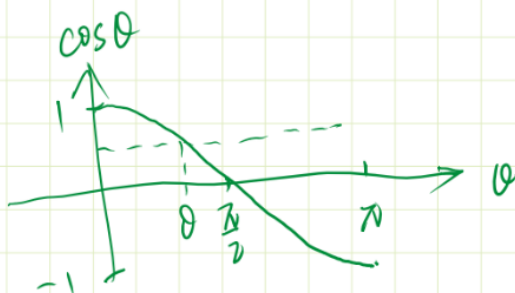
$$|\langle u, v \rangle| \leq \|u\| \|v\| \Leftrightarrow \frac{|\langle u, v \rangle|}{\|u\| \|v\|} \leq 1$$

$$\Leftrightarrow -1 \leq \frac{\langle u, v \rangle}{\|u\| \|v\|} \leq 1$$

Define $\theta \in [0, \pi]$ ($[0^\circ, 180^\circ]$) between u & v by

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

Well defined.



In particular. $u \perp v \Leftrightarrow \langle u, v \rangle = 0 \Leftrightarrow \cos \theta = 0$
 $\Leftrightarrow \theta = \frac{\pi}{2} \text{ (} 90^\circ \text{)}.$